

Asset Data: The Foundation of AI-Driven Autonomous Networks

Unlock autonomous network potential through readily available, reliable asset data

BearingPoint®

ACERNIS

June 2026

Agenda

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Agenda noch einmal anpassen, je nachdem
auf welche Slides wir uns jetzt einigen

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An industry that rose to the challenge

Strong rollout pressure forced mass site deployment

Mobile network operators have faced exceptional challenges in recent years.

The rapid rollout of 4G and 5G required thousands of new sites across urban and rural areas under tight timelines.

At the same time, the industry navigated:

- **Global disruptions:** Pandemic-driven supply chain issues and workforce shortages, combined with a surge in network demand
- **Economic pressure:** Inflation, rising material and energy costs, and geopolitical instability (e.g. war in Ukraine)
- **Complex regulation:** Coordination across authorities, planners, landlords, and subcontractors
- **Ambitious targets:** Government and operator-driven coverage goals with unprecedented delivery speed
- **Talent shortages:** Lack of skilled engineers and technical experts across the supply chain

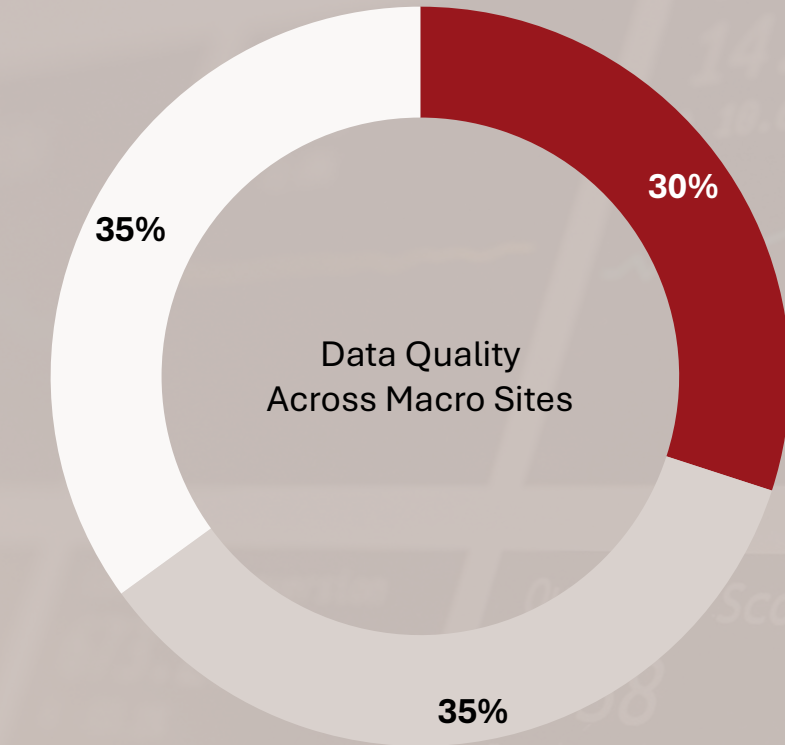
Despite these conditions, Mobile Network Operators and partners delivered at scale, demonstrating strong resilience, adaptability, and commitment.

¹GSMA (2026): *Mobile investment needs in Europe*

Mobile network operators are expected to invest around

€270bn by 2035





- Partially documented (incomplete or outdated)
- Poorly documented (Missing or inconsistent)
- Well documented (Accurate & up-to-date)

The documentation gap: A legacy of speed

Fast rollouts left fragmented and inconsistent asset data

The very conditions that enabled rapid network rollout have, as a natural consequence, left a legacy of uneven asset documentation. Across the industry, **data on macro site infrastructure is often scattered across multiple systems and organizations** and on a significant proportion of sites it is outdated, incomplete, or inconsistent with the physical reality on the ground.

Common deviations between recorded data and as-built conditions include:

- Antenna orientations and mechanical tilt
- Mast positioning and dimensions
- Host building heights and rooftop geometry
- Antenna heights and system technology positioning
- Load-bearing wall and structural element locations
- Skylight positions and dimensions
- Surrounding building heights and distances

These gaps are not the result of negligence. They reflect the reality of working at pace, under resource constraints, across a complex ecosystem of contractors, subcontractors, planning partners, and infrastructure owners - each operating to their own systems, standards, and commercial incentives.

In a low-margin service environment, where partners are remunerated for delivery against specification, the operational focus is understandably on getting sites built and live. **Detailed reconciliation of planned vs. actual conditions has often been deprioritized.** Distributed documentation across multiple parties further limits a single, reliable view of the network.

Why accurate data is now business-critical

A critical enabler for network optimization & automation

Accurate asset data is becoming a critical enabler of next-generation networks. As the industry shifts to optimization and automation, data quality moves from secondary to essential.

High-quality data of physical network assets enables **8+ use cases all the way from coverage prediction to site acceptance.**

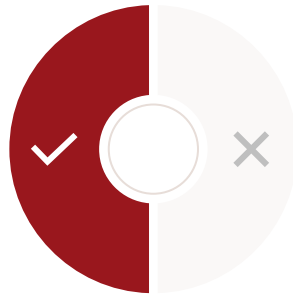
The business impact is significant. Investments in data quality and BIM can reduce site delivery times and enable better investment decisions.

Data sovereignty is becoming strategic. **Owning accurate asset data is as critical as owning infrastructure itself.**



With accurate data:

- Single source of truth
- Data-driven decisions
- Faster rollout cycles (~6 months)
- Full control



Without accurate data:

- Fragmented data
- Reactive decisions
- Slow rollout cycles (>2 years)
- External dependencies



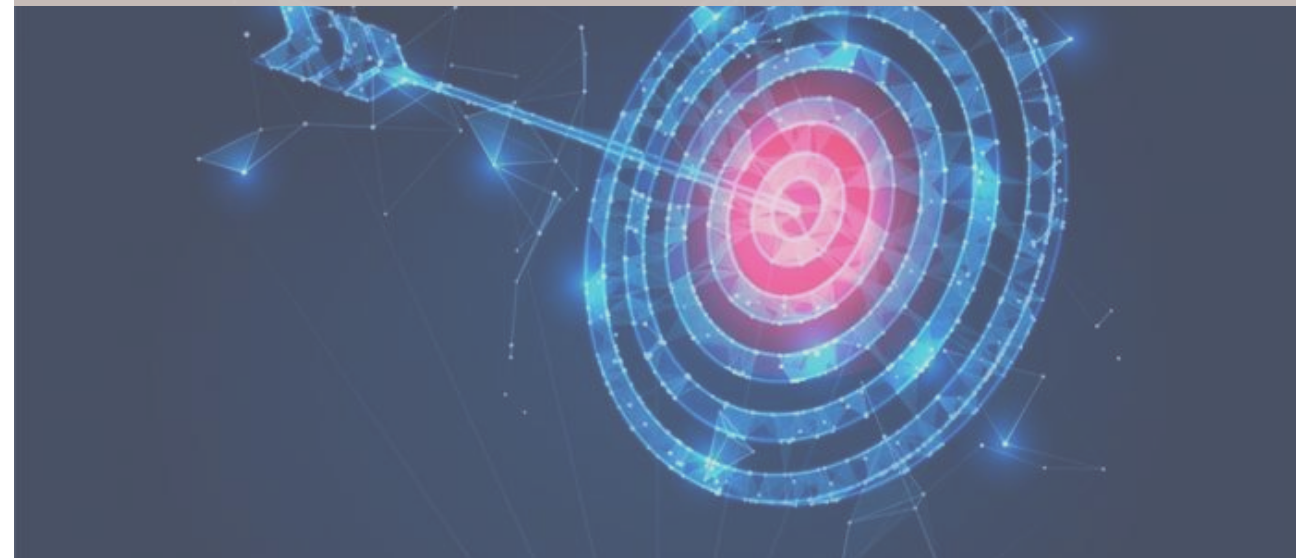
Klaudius Koschella

Director / Head of Digital Processes and Tools,
Network Development Europe - Vodafone

“

Network upgrades should be like throwing a single dart. Set just one parameter and you are certain the execution happens as planned. Accurate, up-to-date data embedded into a well-orchestrated tool landscape is the critical requirement.

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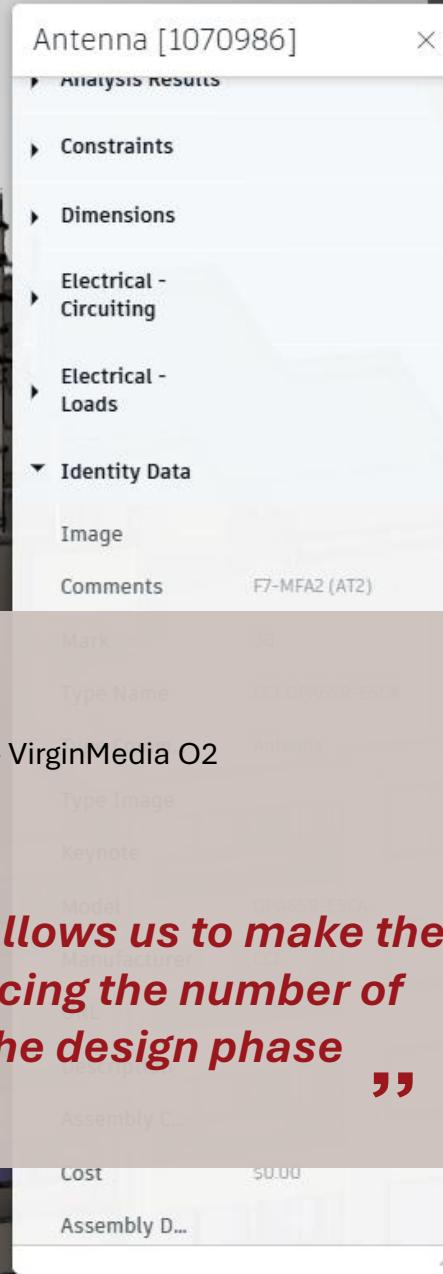
**Thomas Torr**

Mobile Engineering
Design & Build Standards - VirginMedia O2

“

LiDAR scan-based BIM data allows us to make the right decisions early on, reducing the number of iterations drastically during the design phase

”



BIM & site scanning - a single source of truth

Only accurate and trustworthy data ensures reliability

Building Information Modelling (BIM) offers a powerful framework for **representing the full complexity of mobile network infrastructure** in a structured, interoperable format. A well-maintained BIM environment can serve as the single source of truth for site geometry, equipment attributes, structural characteristics, and environmental context feeding multiple downstream applications from a single data model.

However, BIM alone does not resolve the upstream challenge of data accuracy. A BIM model is only as good as the data that populates it. Where input data is estimated, extrapolated, or carried forward from outdated surveys, the model will reflect those limitations.

The most reliable method for establishing a verified, accurate baseline, particularly on sites where historical documentation is incomplete or uncertain, is systematic physical scanning using **laser scanning technologies**. These approaches **eliminate the subjectivity** and variability of manual measurement, capturing precise three-dimensional representations of site conditions that can be directly integrated into BIM workflows.

When combined with a structured **change management process**, one that triggers re-scanning or documentation updates whenever significant modifications are made to a site, systematic scanning provides a sustainable mechanism for maintaining data accuracy over the long term. This creates a virtuous cycle: Verified data at the point of site acceptance, versioned updates as the network evolves, and a **continuously accurate picture of the asset state**.

The Strategic Value of Network Intelligence

Owning asset data: strategic advantage in network decision-making

Data sovereignty is becoming a strategic priority. As network intelligence drives competitive advantage, owning and controlling accurate asset data is as critical as the infrastructure itself.

Data ownership shifts market positioning. A verified, comprehensive view of network assets enables Mobile Network Operators to act from a position of knowledge, credibility, and control, moving from reactive to proactive decision-making.

In practice, this enables:

- **Increasing strategic agility**
- Reducing reliance on external data interpretations
- Faster, more independent internal decisions
- More precise and credible site designs
- Fact-based challenges to contractor assumptions
- Stronger position in co-location negotiations
- Cost savings of e.g. 10-30% for site designs, 10-20% for LoS tests or 30-40% for EMF Assessments

Data ownership = long-term competitive advantage. It increases strategic agility, reduces external dependencies, and enables organizations to shape industry dynamics, not just respond to them.



Vittorio Allegranza

Head of Network Special Projects &
Contract Management - Vodafone

“

It is not about “how do we build more?”, but “how do we build it even smarter?”. That shift depends entirely on data and data structure.

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Cedar Urwyler

Product Owner

Technical Leadership & Steering - Swisscom

“

The vision of autonomous networks starts with well structured, up to date data of our network.

”

How to Close the Data Gap at Scale

From fragmented data to structured, actionable network intelligence

- The shift towards autonomous networks depends on accurate, structured, and continuously maintained asset data as a foundation
- Without a reliable data layer, network operations remain fragmented, reactive, and dependent on assumptions rather than facts
- With high-quality data, operators gain the ability to move towards proactive, data-driven decision-making and unlock network intelligence at scale
- However, achieving this level of data quality requires more than tools or isolated initiatives
- It demands a systematic and scalable approach to how data is captured, structured, and maintained across the entire network lifecycle

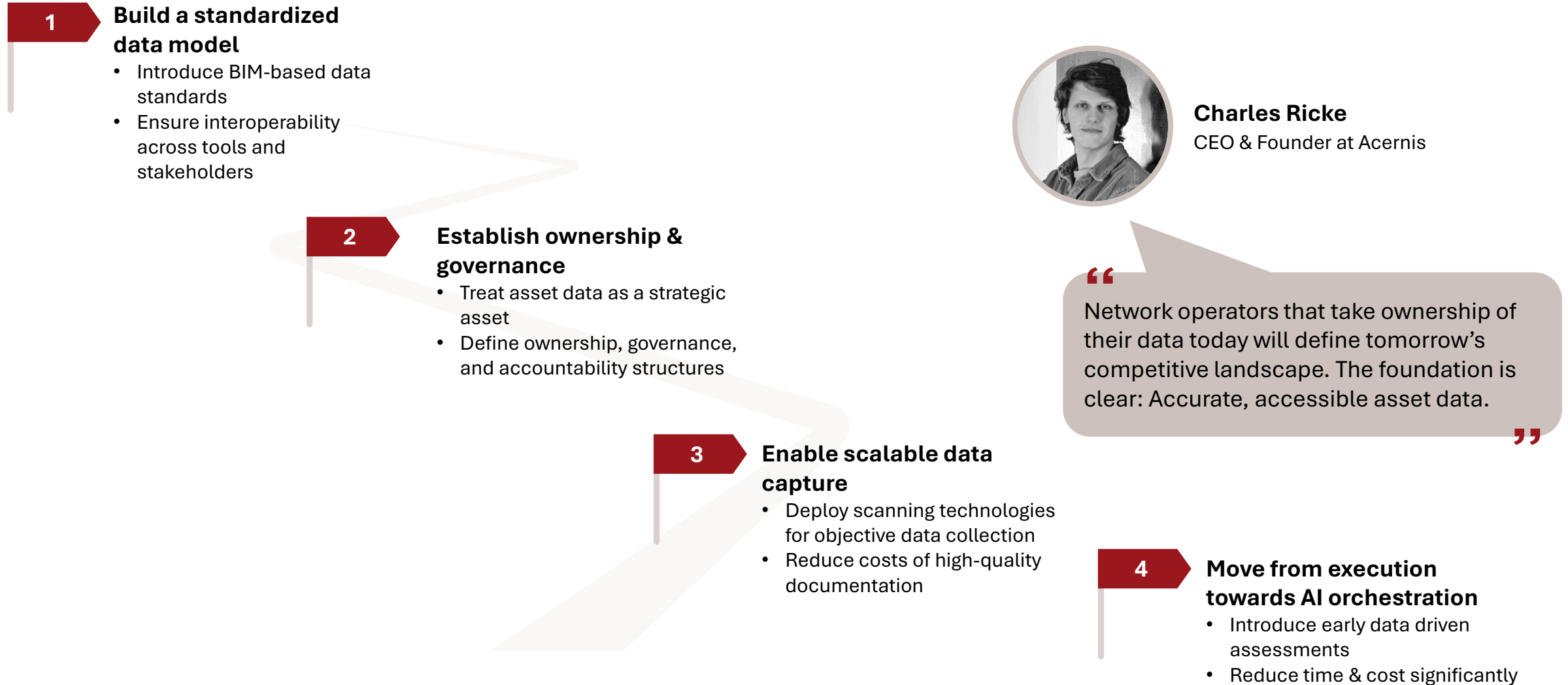


The question is no longer if this transformation is needed - but how to systematically establish and scale a reliable data foundation



A structured path to a reliable data foundation

Establishing high-quality asset data requires clear standards, scalable technologies and strong data ownership



Über BearingPoint

BearingPoint is an independent management and technology consulting firm with European roots and a global reach. We support organizations across industries in addressing complex business challenges by combining deep industry expertise with advanced technological capabilities. Our work spans the full value chain, from strategy and transformation design to implementation and operational execution. The firm operates across three business segments: Consulting, Products, and Capital. Consulting covers the traditional advisory business, offering services across People & Strategy, Customer & Growth, Finance & Risk, Operations, and Technology. In Products, BearingPoint provides clients with IP-based managed services for business-critical processes. Capital covers activities in M&A, ventures, and investments, supporting innovation and long-term value creation. BearingPoint serves many of the world's leading companies and organizations. Its global network of more than 10,000 employees supports clients in over 70 countries, working closely with them to deliver measurable results and sustainable business success. BearingPoint is a certified B Corporation, committed to meeting high standards of social and environmental performance while driving positive impact for clients and society.

More information: www.bearingpoint.com

Über Acernis

Acernis is the AI-powered telecom infrastructure platform, instantly turning network strategy into field-ready deployment instructions.

We help mobile network operators and infrastructure partners accelerate network rollouts by transforming fragmented telecom site data into BIM-based digital twins and AI-readable infrastructure models. Built on top of this foundation, Acernis automates critical rollout workflows, including permitting, rollout planning, contractor coordination, and deployment execution. Our platform enables telecom operators to move from manual, siloed, and slow infrastructure processes toward fully autonomous network operations at scale. With accelerating network demands, operators face increasing complexity, rising capex pressure, and infrastructure programs that routinely run months behind schedule. Acernis provides the data foundation and orchestration layer required to modernize telecom deployment across Europe and beyond. Acernis combines deep telecom domain expertise with advanced AI infrastructure to deliver measurable rollout efficiency and multimillion-euro capex savings. Run and supported by telecom industry experts with 30+ years of hands-on mobile network experience, Acernis is redefining how network infrastructure gets planned, coordinated, and deployed.

More information: www.acernis.fr